

Introduction into Possibilistic Logic

Possibilistic Logic - An Overview by Didier Dubois and Henri Prade

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December 17, 2024



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Possibilistic logic

Possibilistic logic is a weighted logic that handles uncertainty in a qualitative way by adding certainty to classical logic formulas.

- **Applications:** Useable in expert systems, decision-making, and AI for reasoning under uncertainty.

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Possibility distribution

A **possibility distribution** is a mapping:

$$\pi : U \rightarrow S,$$

where:

- ▶ U : Set of states
- ▶ S : Totally ordered scale ($S = [0, 1]$ or finite chains).
- ▶ $\pi(u)$: Degree of possibility (plausibility) of state u .

Example

- ▶ $\pi(u) = 0$: State u is **impossible**.
- ▶ $\pi(u) = 1$: State u is **fully possible**.

Specificity of a Possibility Distribution

A distribution π is **at least as specific** as π' if:

$$\pi(u) \leq \pi'(u), \quad \forall u \in U.$$

Example

- ▶ **Complete knowledge:** $\pi(u_0) = 1$, $\pi(u) = 0$ for all $u \neq u_0$.
- ▶ **Complete ignorance:** $\pi(u) = 1$ for all $u \in U$.

Principle of minimal specificity

It states that any hypothesis not known to be impossible cannot be ruled out. It is a minimal commitment, cautious, information principle.

Possibility measure

$$\Pi(A) = \sup_{u \in A} \pi(u).$$

Evaluates the extent to which A is consistent with π .

Necessity measure

$$N(A) = \inf_{u \notin A} (1 - \pi(u)).$$

Evaluates the extent to which A is certainly implied by π .

Duality Relationship

$$N(A) = 1 - \Pi(A^c).$$

Where A^c is the complement of A .

Possibility: Maxitivity Property

$$\Pi(A \cup B) = \max(\Pi(A), \Pi(B)).$$

Necessity: Minimization Property

$$N(A \cap B) = \min(N(A), N(B)).$$

- Certainty-qualified information: "A is certain to degree α " is modeled by the constraint:

$$N(A) \geq \alpha$$

- Least Specific possibility distribution:

$$\pi_{(A,\alpha)}(u) = \begin{cases} 1 & \text{if } u \in A, \\ 1 - \alpha & \text{otherwise.} \end{cases}$$

Example

- If $\alpha = 1$: $\pi(A, \alpha)$ becomes the characteristic function of A.
- If $\alpha = 0$: $\pi(A, \alpha)$ represents total ignorance.

Example

- ▶ A doctor is trying to diagnose a patient based on symptoms and diseases:
 - ▶ $U = \{\text{flu}, \text{cold}, \text{allergy}\}$.
 - ▶ Possibility distribution π based on symptom severity:

$$\pi(\text{flu}) = 0.9, \quad \pi(\text{cold}) = 0.6, \quad \pi(\text{allergy}) = 0.3.$$

- ▶ The doctor evaluates the possibility and necessity measures for a hypothesis A :

$$A = \{\text{flu}, \text{cold}\}$$

- ▶ Calculation:

$$\Pi(A) = \sup_{u \in A} \pi(u) = \max(\pi(\text{flu}), \pi(\text{cold})) = 0.9.$$

$$N(A) = \inf_{u \notin A} (1 - \pi(u)) = 1 - \pi(\text{allergy}) = 1 - 0.3 = 0.7.$$

- ▶ Interpretation:
 - ▶ Possibility ($\Pi(A)$): Flu or cold is highly possible (0.9).
 - ▶ Necessity ($N(A)$): It is moderately certain (0.7) that the patient has either flu or cold.

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Basic possibilistic logic formula

A basic possibilistic logic formula is represented as:

$$(a, \alpha)$$

where:

- ▶ a : A classical logic formula (e.g., $p \rightarrow q$).
- ▶ $\alpha \in (0, 1]$: Certainty level (minimum necessity degree of a).

Example

- ▶ $(b \rightarrow f, 0.9)$: Birds fly with certainty 0.9.
- ▶ $(p \rightarrow \neg f, 1)$: Penguins do not fly, absolutely certain.

Certainty weakening

$$\text{If } \beta \leq \alpha, \quad (a, \alpha) \vdash (a, \beta)$$

Example

$(\text{It will rain tomorrow}, 0.9) \vdash (\text{It will rain tomorrow}, 0.7)$
If the certainty level decreases, the conclusion still holds.

Modus ponens

$$(\neg a \vee b, \alpha), (a, \alpha) \vdash (b, \alpha), \quad \forall \alpha \in (0, 1]$$

Example

$$(\neg \text{fever} \vee \text{flu}, 0.8), (\text{fever}, 0.8) \vdash (\text{flu}, 0.8)$$

If fever is observed, flu is inferred with certainty 0.8.

Resolution

$$(\neg a \vee b, \alpha), (a \vee c, \alpha) \vdash (b \vee c, \alpha)$$

Example

► Propositions:

- a = "It is raining",
- b = "The ground is wet",
- c = "There is a thunderstorm".

► Resolution:

$$(\neg a \vee b, 0.7), (a \vee c, 0.7) \vdash (b \vee c, 0.7)$$

- Interpretation: "The ground is wet or there is a thunderstorm" holds with certainty 0.7.

Weakest link resolution

$$(\neg a \vee b, \alpha), \quad (a \vee c, \beta) \vdash (b \vee c, \min(\alpha, \beta))$$

Example

- Premises:

$$(\neg a \vee b, 0.7), \quad (a \vee c, 0.5).$$

- Inference:

$$(\neg a \vee b, 0.7), \quad (a \vee c, 0.5) \vdash (b \vee c, \min(0.7, 0.5)).$$

- Interpretation: "The ground is wet or there is a thunderstorm" is true with certainty 0.5, limited by the weaker premise.

Formula weakening

If $a \vdash b$ in classical logic, then $(a, \alpha) \vdash (b, \alpha)$.

Example

► Premise:

$(a, 0.8)$ ("It is raining with certainty 0.8").

► Conclusion:

$(b, 0.8)$ ("The ground is wet with certainty 0.8").

Knowledge base (Γ)

$$\Gamma = \{(a_i, \alpha_i), i = 1, \dots, m\}$$

Properties

- ▶ Nested structure: $\Gamma_\alpha \subseteq \Gamma_\beta$ if $\beta \leq \alpha$.
- ▶ Formulas in Γ_α contribute to inferences at certainty level α .

Knowledge base (Γ_α)

The knowledge base Γ is divided into nested level cuts:

$$\Gamma_\alpha = \{(a_i, \alpha_i) \in \Gamma \mid \alpha_i \geq \alpha\}$$

Knowledge base (Γ_α)

Classical logic knowledge base derived from Γ , consisting of only the formulas a_i (without their certainty levels)

$$\Gamma^* = \{a_i \mid (a_i, \alpha_i) \in \Gamma\}$$

Inference

$$\Gamma \vdash (a, \alpha) \iff \Gamma_\alpha \vdash (a, \alpha) \iff (\Gamma_\alpha)^* \vdash a,$$

- ▶ A possibilistic inference (a, α) depends only on formulas with certainty levels $\geq \alpha$.
- ▶ These formulas can be reduced to classical logic formulas for inference.

Example

- ▶ Knowledge Base:

$$\Gamma = \{(a_1, 0.9), (a_2, 0.7), (a_3, 0.5)\}.$$

- ▶ Inference for $\alpha = 0.7$:

$$(\Gamma_\alpha)^* \vdash a_2 \implies \Gamma_\alpha \vdash (a_2, 0.7) \implies \Gamma \vdash (a_2, 0.7).$$

Inconsistency level of Γ

$$\text{inc-l}(\Gamma) = \max\{\alpha \mid \Gamma \vdash (\perp, \alpha)\}.$$

Properties

- Formulas with certainty levels $> \text{inc-l}(\Gamma)$ are safe from inconsistency.

$$\text{inc-l}(\{(a_i, \alpha_i) \mid (a_i, \alpha_i) \in \Gamma \text{ and } \alpha_i > \text{inc-l}(\Gamma)\}) = 0.$$

- If $(\Gamma)^*$ (the classical counterpart of Γ) is consistent, then:

$$\text{inc-l}(\Gamma) = 0.$$

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- ▶ Default Reasoning
- ▶ Belief Revision and Information Fusion
- ▶ Decision Under Uncertainty:
- ▶ Handling Uncertainty in Information Systems
- ▶ Logic Programming
- ▶ Applications in Diagnosis and Recognition

- ▶ Default rule:

$$\Pi(a \wedge b) > \Pi(a \wedge \neg b)$$

- $a \rightarrow b$: "If a , then b is more plausible than $\neg b$."

- ▶ In terms of necessity:

$$N(b|a) = 1 - \Pi(\neg b|a) > 0$$

- ▶ Comparative relation:

$$a \wedge b >_{\Pi} a \wedge \neg b$$

► Default Rules:

$d_1 : b \rightarrow f$ ("Birds fly"),

$d_2 : p \rightarrow \neg f$ ("Penguins do not fly"),

$d_3 : p \rightarrow b$ ("Penguins are birds").

► Constraints:

$C_1 : b \wedge f >_{\square} b \wedge \neg f$ (birds fly).

$C_2 : p \wedge \neg f >_{\square} p \wedge f$ (penguins don't fly).

$C_3 : p \wedge b >_{\square} p \wedge \neg b$ (penguins are birds).

► Principle of Minimal Specificity: Assigns possibility levels to worlds while satisfying these constraints.

► Worlds (Ω):

$$\Omega = \{\omega_0, \omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7\},$$

where each ω_i represents a unique combination of b, f, p (bird, fly, penguin), e.g:

$$\begin{aligned} \omega_0 : \neg b \wedge \neg f \wedge \neg p, \quad \omega_1 : \neg b \wedge \neg f \wedge p, \quad \omega_2 : \neg b \wedge f \wedge \neg p, \quad \omega_3 : \neg b \wedge f \wedge p, \quad \omega_4 : \\ b \wedge \neg f \wedge \neg p, \quad \omega_5 : b \wedge \neg f \wedge p, \quad \omega_6 : b \wedge f \wedge \neg p, \quad \omega_7 : b \wedge f \wedge p \end{aligned}$$

► Constraints:

$$C_1 : \max(\pi(\omega_6), \pi(\omega_7)) > \max(\pi(\omega_4), \pi(\omega_5)),$$

$$C_2 : \max(\pi(\omega_5), \pi(\omega_1)) > \max(\pi(\omega_3), \pi(\omega_7)),$$

$$C_3 : \max(\pi(\omega_5), \pi(\omega_7)) > \max(\pi(\omega_1), \pi(\omega_3)).$$

- ▶ Default Rule $d_1 : b \rightarrow f$ ("Birds fly"):
 - ▶ Makes worlds where birds fly ($b \wedge f$) more plausible than those where birds don't fly ($b \wedge \neg f$).
 - ▶ Affects:
 ω_6 (bird flies) vs. ω_4, ω_5 (bird doesn't fly).
- ▶ Default Rule $d_2 : p \rightarrow \neg f$ ("Penguins don't fly"):
 - ▶ Makes worlds where penguins don't fly ($p \wedge \neg f$) more plausible than those where penguins fly ($p \wedge f$).
 - ▶ Affects:
 ω_5 (penguin doesn't fly) vs. ω_7, ω_3 (penguin flies).
- ▶ *Default Rule $d_3 : p \rightarrow b$ ("Penguins are birds"):
 - ▶ Ensures penguins are considered birds, influencing how penguin-related worlds (p) are ranked.

- ▶ Partition: Worlds are ranked based on constraints:

$$\{\omega_0, \omega_2, \omega_6\} >_{\Pi} \{\omega_4, \omega_5\} >_{\Pi} \{\omega_1, \omega_3, \omega_7\}.$$

- ▶ m is the number of distinct plausibility levels (possibility degrees) assigned to the worlds in Ω .
- ▶ Possibility Levels:

$$\pi(\omega_0) = \pi(\omega_2) = \pi(\omega_6) = 1, \quad \pi(\omega_4) = \pi(\omega_5) = 2/3, \quad \pi(\omega_1) = \pi(\omega_3) = \pi(\omega_7) = 1/3.$$

- ▶ Necessity Measure ($N(A)$):

$$N(A) = \inf_{u \notin A} (1 - \pi(u)).$$

- ▶ Computation:

- ▶ $N(\neg p \vee \neg f)$:

$$N(\neg p \vee \neg f) = \min\{1 - \pi(\omega_3), 1 - \pi(\omega_7)\} = \min\{1 - 1/3, 1 - 1/3\} = 2/3.$$

- ▶ $N(\neg b \vee f)$:

$$N(\neg b \vee f) = \min\{1 - \pi(\omega_4), 1 - \pi(\omega_5)\} = \min\{1 - 2/3, 1 - 2/3\} = 1/3.$$

- ▶ $N(\neg p \vee b)$:

$$N(\neg p \vee b) = \min\{1 - \pi(\omega_1), 1 - \pi(\omega_3)\} = \min\{1 - 1/3, 1 - 1/3\} = 2/3.$$

- ▶ Resulting Possibilistic Logic Base (K):

$$K = \{(\neg p \vee \neg f, 2/3), (\neg p \vee b, 2/3), (\neg b \vee f, 1/3)\}.$$

- Initial Information (Tweety is a Bird):

$$F = \{(b, 1)\} \quad (\text{Tweety is a bird with certainty } 1).$$

- Apply the weakest link resolution rule:

$$K \cup \{(b, 1)\} \vdash (f, 1/3).$$

- Conclusion: If Tweety is a bird, we conclude that Tweety flies with certainty $\frac{1}{3}$.

- New Information (Tweety is a Penguin):

$$F = \{(b, 1), (p, 1)\} \quad (\text{Tweety is a bird and a penguin with certainty } 1).$$

- Apply the weakest link resolution rule:

$$K \cup F \vdash (\perp, 1/3).$$

- Conclusion: The knowledge base becomes inconsistent with certainty $\frac{1}{3}$.

- Final Conclusion (Tweety is a Penguin): - Despite inconsistency, we can still infer:

$$K \cup F \vdash (\neg f, 2/3).$$

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- ▶ Possibilistic logic provides a robust framework for reasoning under uncertainty.
- ▶ Key features:
 - ▶ Certainty levels for qualitative inference.
 - ▶ Compatibility with classical logic.
 - ▶ Handling of conflicting information.
- ▶ Applications:
 - ▶ Medical diagnosis, decision-making, and knowledge representation.

7 Appendix

This is an appendix.

- ▶ Page numbers start from the beginning.